

Microeconomic Analysis

Topics on Information Economics

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Topics on Information Economics

I Hidden Action in Principal-Agent Problems

II Hidden Information in Principal-Agents Problems

Introduction

In Core Micro you saw that **asymmetric information** usually causes inefficiencies in markets and interactions.

Asymmetric information might occur **before** contracting:

- **Adverse Selection:** *market for used-cars*

And you saw possible market reactions to adverse selection:

- ▶ Signalling: *education and job market*
- ▶ Screening: *Insurance*

Asymmetric information might occur **after** contracting:

- **Principal-Agent problems.**

Principal Agent Problems

These are situations in which:

- Symmetric information *at the time of contracting*
- asymmetries arise *after contracting*.

One individual, the **principal**, hires another, the **agent**, to take some action on her behalf.

We will analyse two different sort of problems:

- 1 **Hidden Action:** The principal cannot observe the how much effort the agent exerts. *Moral Hazard*.
- 2 **Hidden Information:** The agent discovers the profitability of opportunities that the principal cannot observe.

We will analyse these problems in terms of an owner-manager relationship. But it applies to many settings:

Insurance companies and insured individuals, manufacturers and their distributors, an employer and its workforce, a bank and its borrowers.

Principal-Agent Problems: Hidden Action

A **risk neutral** principal hires a **risk averse** agent for a one-time project. The agent chooses an action that will affect the project's profit.

We will call that action “**effort**” and will focus on the simple case of two possible levels of effort: $e = e_L$ or e_H ; $e_L < e_H$.

The effort is unobservable to the principal, and it cannot be perfectly deducible from the observation of the profits of the project π .

The profits π are distributed on $[\pi_L, \pi_H]$, and the distribution depends on the effort e .

Assumption

The distribution of π conditional on e_H **first-order stochastically dominates** the distribution of π conditional on e_L .

Principal-Agent Problems: Hidden Action

As a recap from last year:

Definition: FOSD

The distribution $F(\cdot)$ **first order stochastically dominates** $G(\cdot)$ if, for every nondecreasing function $u : \mathbb{R} \rightarrow \mathbb{R}$, we have

$$E[u(x) \mid F] \geq E[u(x) \mid G]$$

In particular, taking $u(x) = x$, this implies that $E[x \mid F] > E[x \mid G]$, or in our setting:

$$E[\pi \mid e_H] > E[\pi \mid e_L]$$

A necessary and sufficient condition for F to FOSD G :

$$F(x) \leq G(x) \quad (\text{with '<' for some } x)$$

or in our setting:

$$F(\pi \mid e_H) \leq F(\pi \mid e_L) \quad \forall \pi \in [\pi_L, \pi_H]$$

Principal Agent Problems- Hidden Action

Agent's Preferences:

Agent's utility from wage and effort:

$$u(w, e) = v(w) - g(e)$$

with $v' > 0$, $v'' \leq 0$, $g' > 0$.

- $v'' \leq 0$: Agent is risk-averse
- $g' > 0$: higher level of effort is more costly to the agent.

Agent has *ex ante* reservation utility $\bar{u}$.

Principal's Preferences:

Principal's utility from wage w and effort e is the expected net profit :

$$E[\pi - w \mid e]$$

- Principal is risk-neutral

Principal Agent Problems- Hidden Action

Effort e is non-observable by principal (or, equivalently, non-contractible).

Wage w can be conditioned on profit π .

Principal's problem:

Choose $e \in \{e_L, e_H\}$ and the wage scheme $w(\pi)$ to maximise

$$E [\pi - w(\pi) \mid e]$$

such that the agent wishes to *participate* (**Individual Rationality**)

$$E [v(w(\pi)) \mid e] - g(e) \geq \bar{u}$$

and that the agent wishes to *choose* e (**Incentive Compatibility**), i.e. e maximises

$$E [v(w(\pi)) \mid e] - g(e)$$

Principal Agent Problems- Hidden Action

We solve this problem in two stages:

- 1 for each e_i , find the optimal $w_i(\pi)$ that implements e_i ;

$$\min_{w(\pi)} E[w(\pi)] \quad s.t.$$

$$E[v(w(\pi)) | e_i] - g(e_i) \geq \bar{u}$$

$$e_i = \arg \max_e E[v(w(\pi)) | e] - g(e)$$

- 2 find the optimal e^* :

$$e^* = \arg \max E[\pi - w_i(\pi) | e_i]$$

Trade-off: risk-sharing vs. incentives.

Benchmark I: Observable effort (contractible)

When effort is observable (and contractible), a contract specifies the agent's effort and the wage scheme $w(\pi)$:

Incentives are not a concern.

Ignore the **Incentive constraint**;

For stage 1: for each e ,

$$\min_{w(\pi)} E[w(\pi) | e] \quad \text{such that} \quad E[v(w(\pi)) | e] - g(e) \geq \bar{u}$$

The **Participation constraint** will bind:

If $E[v(w(\pi)) | e] - g(e) > \bar{u}$, we can decrease the wage while satisfying the constraint and hence achieve a lower expected wage.

Benchmark I: Observable effort (contractible)

So:

$$\min_{w(\pi)} E[w(\pi) | e] \quad \text{such that} \quad E[v(w(\pi)) | e] - g(e) = \bar{u}$$

Consider for simplicity a discrete distribution of π , then

$$E[w(\pi) | e] = \sum_{\pi} w(\pi) Pr[\pi | e],$$

$$E[v(w(\pi)) | e] = \sum_{\pi} v(w(\pi)) Pr[\pi | e]$$

We construct the Lagrangian:

$$L(\{w(\pi)\}_{\pi}; \gamma) = \sum_{\pi} w(\pi) Pr[\pi | e] - \gamma \left(\sum_{\pi} v(w(\pi)) Pr[\pi | e] - g(e) - \bar{u} \right)$$

For each π , the FOC with respect to $w(\pi)$ is:

$$Pr[\pi | e] - \gamma v'(w(\pi)) Pr[\pi | e] = 0$$

so

$$\gamma = \frac{1}{v'(w(\pi))}$$

Benchmark I: Observable effort (contractible)

$$\gamma = \frac{1}{v'(w(\pi))} > 0$$

In particular, this means that for all π , $v'(w(\pi)) = \frac{1}{\gamma}$, or in other words if $v'' < 0$, the optimal wage scheme is **constant**.

(If $v'' = 0$ there are multiple solutions.)

Intuition:

The agent is risk averse, using a variable wage scheme introduces risk, and hence on average it is more costly to induce the reservation utility.

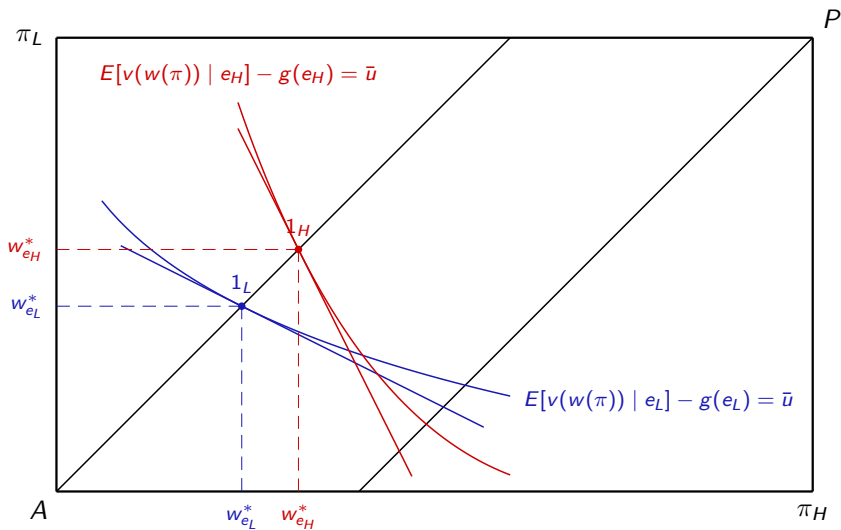
Therefore $w(\pi)$ will be fixed at $w_{e_H}^*$ or $w_{e_L}^*$ such that

$$v(w_{e_H}^*) - g(e_H) = \bar{u} \quad \text{and} \quad v(w_{e_L}^*) - g(e_L) = \bar{u}$$

Note that $w_{e_H}^* > w_{e_L}^*$.

Benchmark I: Observable effort (contractible)

Illustration for the case of two values of π : π_L, π_H .



Benchmark I: Observable effort (contractible)

Stage 2: This is a simple trade-off between expected profits and wages:

$$\max_{e \in \{e_L, e_H\}} \{E[\pi \mid e_L] - w_{e_L}^*, E[\pi \mid e_H] - w_{e_H}^*\}$$

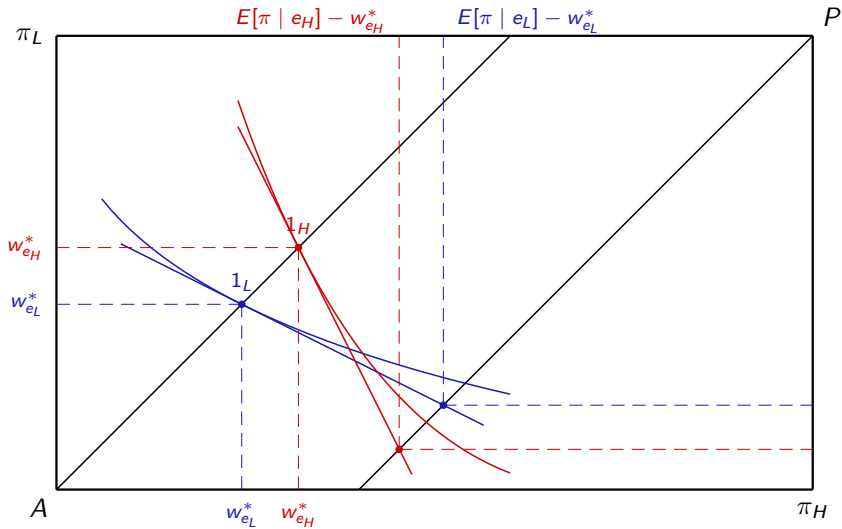
This contract achieves two goals:

- Efficient effort choice by the agent.
- No risk to the agent

Full information Payoffs:

- The agent gets her reservation utility $\bar{u}$
- Principal gets: $E[\pi \mid e^*] - v^{-1}(\bar{u} + g(e^*))$.

Benchmark I: Observable effort (contractible)



In this case: $e^* = e_H$

Benchmark II: Unobservable effort, Agent risk-neutral

If effort is unobservable but the agent is risk-neutral:

Risk-sharing is not a concern.

Take $v(w) \equiv w$.

If the effort was observable, the optimal effort e^* would maximise:

$$E[\pi - w_e^* | e] = E[\pi | e] - g(e) - \bar{u}$$

which coincides with the effort that maximises social surplus, since the expected wage, $E[w_e^*(\pi) | e]$, is merely a transfer between principal and agent.

Can the principal achieve her **full information** payoff when the effort is unobservable?

Consider the following contract:

The principal *sells the project* to the agent at price α .

Benchmark II: Unobservable effort, Agent risk-neutral

Selling the project at price α is equivalent to setting a wage scheme:

$$w(\pi) = \pi - \alpha$$

In that case the agent chooses e to maximise

$$E[\pi - \alpha \mid e] - g(e)$$

The agent's **incentives** are aligned to social surplus maximisation (i.e. the agent will choose effort e^*);

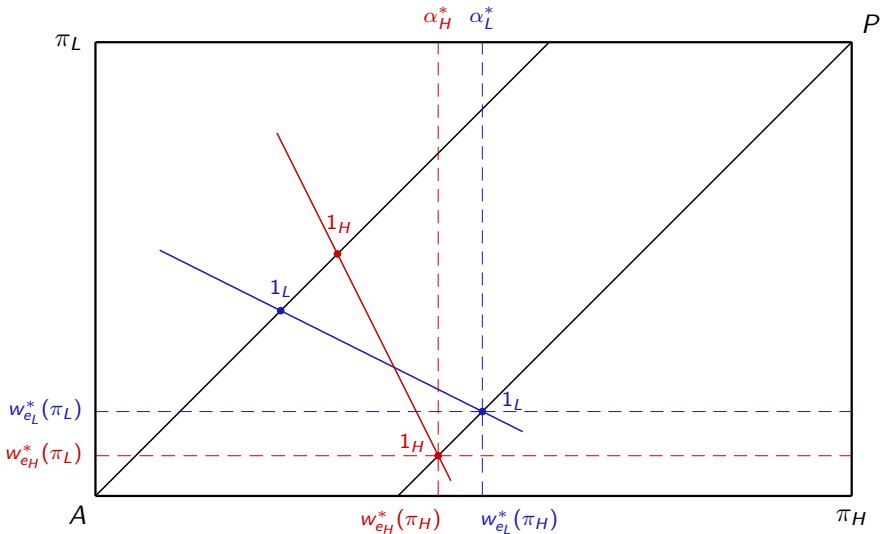
And the principal can choose α optimally, so that the agent's **Participation constraint** binds:

$$\alpha^* = E[\pi \mid e^*] - g(e^*) - \bar{u}$$

Hence the principal (and the agent) get the same payoff as when the effort was observable:

- Agent gets $E[\pi \mid e^*] - (E[\pi \mid e^*] - g(e^*) - \bar{u}) - g(e^*) = \bar{u}$.
- Principal gets: $\alpha^* = E[\pi \mid e^*] - g(e^*) - \bar{u}$

Benchmark II: Unobservable effort, Agent risk-neutral



In this case: $e^* = e_H$

Unobservable effort, Agent risk-averse.

When effort is unobservable and the agent is risk averse:

Trade off: incentives for high effort can be provided only at the cost of risk to the agent.

Stage 1:

To implement e_L , it is sufficient for the principal to offer the fixed wage $w_{e_L}^*$ that satisfies

$$v(w_{e_L}^*) - g(e_L) = \bar{u}$$

- the Participation constraint binds;
- facing a fixed wage, the agent strictly prefers to choose e_L rather than e_H .

Unobservable effort, Agent risk-averse.

To implement e_H : find the optimal $w(\pi)$ that minimises $E[w(\pi) \mid e_H]$ subject to

$$E[v(w(\pi)) \mid e_H] - g(e_H) \geq \bar{u}$$

$$E[v(w(\pi)) \mid e_H] - g(e_H) \geq E[v(w(\pi)) \mid e_L] - g(e_L)$$

We construct the Lagrangian:

$$\begin{aligned} L(\{w(\pi)\}_\pi; \gamma, \mu) = & \sum_\pi w(\pi) Pr[\pi \mid e_H] - \gamma (\sum_\pi v(w(\pi)) Pr[\pi \mid e_H] - g(e_H) - \bar{u}) \\ & - \mu (\sum_\pi v(w(\pi)) (Pr[\pi \mid e_H] - Pr[\pi \mid e_L]) - (g(e_H) - g(e_L))) \end{aligned}$$

The Kuhn-Tucker FOCs imply that for every π , the wage $w(\pi)$ satisfies:

$$Pr[\pi \mid e_H] - \gamma v'(w(\pi)) Pr[\pi \mid e_H] - \mu v'(w(\pi)) (Pr[\pi \mid e_H] - Pr[\pi \mid e_L]) = 0$$

or

$$\frac{1}{v'(w(\pi))} = \gamma + \mu \left[1 - \frac{Pr[\pi \mid e_L]}{Pr[\pi \mid e_H]} \right]$$

Unobservable effort, Agent risk-averse.

$$\frac{1}{v'(w(\pi))} = \gamma + \mu \left[1 - \frac{Pr[\pi | e_L]}{Pr[\pi | e_H]} \right]$$

Binding constraints

In any solution of this problem, both $\gamma > 0$ and $\mu > 0$

Proof:

- Suppose $\mu = 0$, then we will have a constant wage schedule. But then the agent will always choose e_L since it leads to same constant wage and is less costly.

Unobservable effort, Agent risk-averse.

- Suppose $\gamma = 0$. Because the distribution of π conditional on e_H FOSD the distribution of π conditional on e_L , for some low values of π we must have:

$$\Pr[\pi \mid e_L] > \Pr[\pi \mid e_H]$$

But then for those π we would have

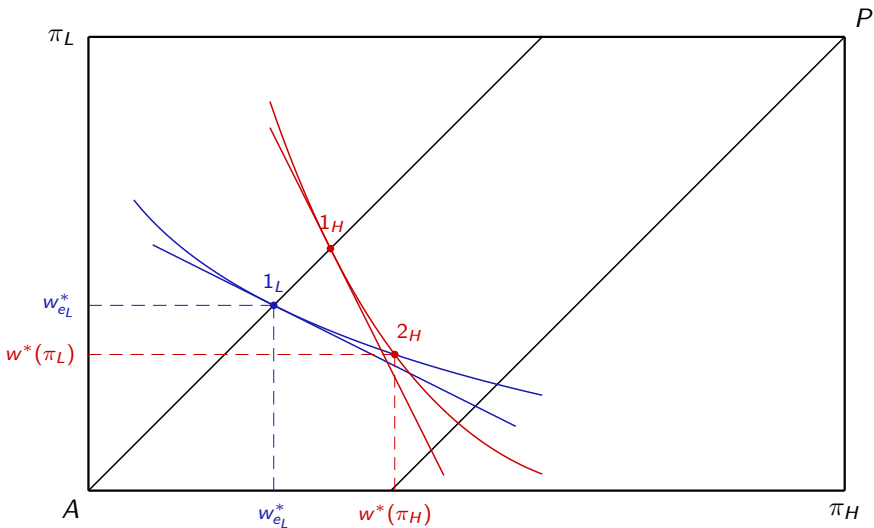
$$\frac{1}{v'(w(\pi))} = \mu \left[1 - \frac{\Pr[\pi \mid e_L]}{\Pr[\pi \mid e_H]} \right] \leq 0$$

which implies $v'(w(\pi)) < 0$ (since $\mu > 0$) which is a contradiction.

Hence both the participation constraint and the incentive constraint are binding.

Unobservable effort, Agent risk-averse.

Illustration for the case of two values for π : π_L, π_H .



Unobservable effort, Agent risk-averse.

In order to implement e_H , the expected wage is higher under non-observability:

For the risk averse agent to accept the gamble, the expected utility needs to be at least $\bar{u}$, which means that the **certainty equivalence** must be at least $w_{e_H}^*$.

The risk premium associated to the wage scheme, is called the **Agency cost**.

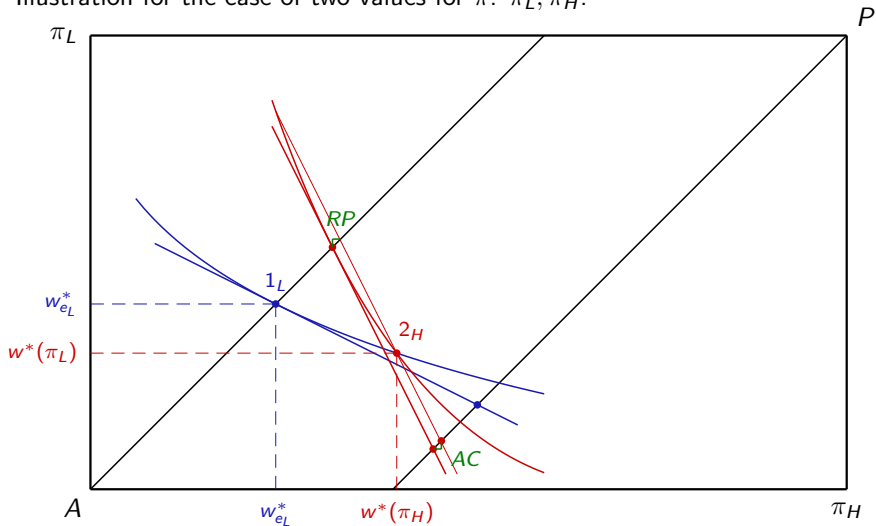
$$\text{Agency cost} = E[w \mid e_H] - w_{e_H}^*$$

Stage 2:

$$\max_{e \in \{e_L, e_H\}} \{E[\pi \mid e_L] - w_{e_L}^*, E[\pi - w \mid e_H]\}$$

Unobservable effort, Agent risk-averse.

Illustration for the case of two values for π : π_L, π_H .



In this particular case it is still optimal to induce e_H

Unobservable effort, Agent risk-averse.

Summary:

- The optimal compensation scheme that implements e_L is the same as when effort is observable: *Fixed* wage and the agent's utility is $\bar{u}$.
- The optimal compensation scheme that implements e_H also gives the agent an *expected* utility of $\bar{u}$, but involves a *higher* expected wage than is required when effort is observable.
- Whenever the optimal effort level with observable effort is e_H , there is a welfare loss (to the principal) under non-observability.

Unobservable effort, Agent risk-averse.

Can we say something more about the optimal wage scheme?

Return to the K-T FOC:

$$\frac{1}{v'(w(\pi))} = \gamma + \mu \left[1 - \frac{\Pr[\pi | e_L]}{\Pr[\pi | e_H]} \right]$$

Therefore, if π_1 and π_2 are such that

$$\Pr[\pi_1 | e_L] / \Pr[\pi_1 | e_H] = \Pr[\pi_2 | e_L] / \Pr[\pi_2 | e_H] \quad \text{then} \quad w(\pi_1) = w(\pi_2)$$

and if

$$\Pr[\pi_1 | e_L] / \Pr[\pi_1 | e_H] > \Pr[\pi_2 | e_L] / \Pr[\pi_2 | e_H] \quad \text{then} \quad w(\pi_1) < w(\pi_2)$$

Unobservable effort, Agent risk-averse.

Moreover, if $\hat{w}$ is the fixed wage such that $(1/v'(\hat{w})) = \gamma$:

$$w(\pi) > \hat{w} \quad \text{if} \quad \frac{\Pr[\pi \mid e_L]}{\Pr[\pi \mid e_H]} < 1 \quad \text{and} \quad w(\pi) < \hat{w} \quad \text{if} \quad \frac{\Pr[\pi \mid e_L]}{\Pr[\pi \mid e_H]} > 1$$

So, if a realisation of π is relatively *more* likely to occur under e_H than under e_L , then the optimal reward scheme pays *more*;

and if a realisation of π is relatively *less* likely to occur under e_H than under e_L , then the optimal reward scheme pays *less*.

Thus, if

$$\frac{\Pr[\pi \mid e_H]}{\Pr[\pi \mid e_L]} \text{ is increasing in } \pi \text{ (Monotone Likelihood Ratio Property),}$$

then the wage schedule $w(\pi)$ will be increasing in π .

Note: FOSD $\nRightarrow$ MLRP.

Example: Linear Compensation schemes

Consider a risk neutral owner and a risk averse manager. The manager has preferences over the mean and variance of her income w and her effort e as follows:

$$Eu(w, e) = E[w] - \phi \text{Var}[w] - g(e)$$

With $g', g'', g''' > 0$ for $e > 0$ and $\lim_{e \rightarrow \infty} g'(e) = \infty$.

Conditional on effort level e , the profit π is **normally distributed** with mean e and variance σ^2 .

Restrict attention to linear wage schemes $w(\pi) = a + b\pi$.

- 1 Show that the manager's expected utility when exerting effort e is

$$a + be - \phi\sigma^2 b^2 - g(e)$$

- 2 Find the optimal contract when effort is observable (and verifiable), assuming that the manager's reservation utility is normalised to 0.
- 3 When effort is not observable, what is the optimal linear wage schedule? Show that $0 < b < 1$ and interpret.
- 4 What is the effect of a change in ϕ or σ^2 ?

Example: Linear Compensation schemes

- 1 The expected utility when $w(\pi) = a + b\pi$ is:

$$\begin{aligned}Eu(w, e) &= E[w] - \phi \text{Var}[w] - g(e) \\&= E[a + b\pi \mid e] - \phi \text{Var}[a + b\pi \mid e] - g(e) \\&= a + bE[\pi \mid e] - \phi b^2 \text{Var}[\pi \mid e] - g(e) \\&= a + be - \phi b^2 \sigma^2 - g(e)\end{aligned}$$

- 2 Intuitively, as the owner is risk neutral and the manager is risk averse, the optimal contract when effort is observable (and verifiable) will result in a fixed wage for the manager ($b^* = 0$) who then achieves her reservation utility, i.e. $w(e) = a = g(e)$.

The owner then chooses the effort level that maximises expected profit:

$$\max_e \{E[\pi \mid e] - w(e)\} = \max_e \{e - g(e)\}$$

and the FOC (which is necessary and sufficient) gives $g'(e^*) = 1$.

Thus $e^* = (g')^{-1}(1)$, and $w = a^* = g(e^*)$.

Example: Linear Compensation schemes

- 3 When effort is not observable, we first find the least cost method of implementing a given level of effort:

Given e , the optimal compensation scheme solves:

$$\min_{a,b} E[a + b\pi \mid e] \quad s.t.$$

$$(IR) \quad a + be - \phi\sigma^2 b^2 - g(e) \geq 0$$

$$(IC) \quad e = \arg \max a + be - \phi\sigma^2 b^2 - g(e)$$

(IC) implies that e maximises $be - g(e)$ and the FOC (which is necessary and sufficient) gives $g'(e) = b$. So b is determined uniquely given e .

(IR) will bind (why?). So $a = g(e) + \phi\sigma^2 g'(e)^2 - g'(e)e$

Example: Linear Compensation schemes

We have found the optimal wage scheme to implement e . The expected wage, given e is:

$$\begin{aligned} E[w(\pi) \mid e] &= a + be \\ &= g(e) + \phi\sigma^2 g'(e)^2 - g'(e)e + g'(e)e \\ &= g(e) + \phi\sigma^2 g'(e)^2 \end{aligned}$$

Now the owner chooses the effort level which solves

$$\max_e \{E[\pi - w(\pi) \mid e]\} = \max_e \{e - g(e) - \phi\sigma^2 g'(e)^2\}$$

and the FOC (which is necessary and sufficient) gives

$$1 - g'(\hat{e}) - 2\phi\sigma^2 g'(\hat{e})g''(\hat{e}) = 0$$

And so the optimal effort, $\hat{e}$, satisfies:

$$g'(\hat{e}) = \frac{1}{1 + 2\phi\sigma^2 g''(\hat{e})}$$

Example: Linear Compensation schemes

$$g'(\hat{e}) = \frac{1}{1 + 2\phi\sigma^2 g''(\hat{e})}$$

In particular $\hat{b} = g'(\hat{e}) \in (0, 1)$:

- ▶ Incentives are not fully aligned ($\hat{b} < 1$) - the first best effort satisfied $g'(e^*) = 1$.
- ▶ In particular since $g'' > 0$: $\hat{e} < e^*$.

4 How does $\hat{e}$ change with ϕ or σ^2 ?

We can use the IFT on the equation:

$$H(e, \phi, \sigma^2) = g'(e)(1 + 2\phi\sigma^2 g''(e)) = 1$$

The function H is C^1 and

$$\frac{\partial H}{\partial e}(\hat{e}, \phi, \sigma^2) \equiv g''(\hat{e})(1 + 2\phi\sigma^2 g''(\hat{e})) + 2\phi\sigma^2 g'(\hat{e})g'''(\hat{e}) > 0$$

Example: Linear Compensation schemes

Therefore $\hat{e}$ is an implicit function of ϕ and σ^2 and:

$$\frac{\partial \hat{e}}{\partial \phi} = \frac{\partial \hat{e}}{\partial \sigma^2} = -\frac{g'(\hat{e})2\sigma^2 g''(\hat{e})}{g''(\hat{e})(1 + 2\phi\sigma^2 g''(\hat{e})) + 2\phi\sigma^2 g'''(\hat{e})} < 0$$

In particular $\hat{b} = g'(\hat{e})$ decreases in ϕ and in σ^2 ;

Intuition: the manager is more risk averse (or faces a higher variance) and this is reflected in the smaller variable part of his compensation.

Unobservable effort, Agent risk-averse.

Summary:

- The optimal linear compensation scheme when the agent is risk neutral is to *sell the project*, $b = 1$, and $a = \bar{u} - e^* + g(e^*)$ where $g'(e^*) = 1$.
- If the agent is risk averse but the effort is observable, then $b = 0$ and $a = g(e^*)$ where $g'(e^*) = 1$.
- If the agent is risk averse and effort is unobservable, then the optimal linear contract:
 - ▶ involves positive (but not fully aligned) incentives $b \in (0, 1)$.
 - ▶ induces inefficient level of effort: $\hat{e} < e^*$.
 - ▶ The level of effort decreases with risk aversion or the with the noise of the signal.
- *Extensions:* Multiple actions, Additional signals, Repeated interaction, Multi-tasking, ...